

BAUDETTE, MN, USA

WMO: 727476

Lat: 48.717N

Lon: 94.600W

Elev: 330

StdP: 97.43

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 94961

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.3	-28.6	-33.8	0.2	-30.1	-31.7	0.2	-27.7	11.0	-12.9	9.9	-13.8	1.7	310	0.628

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	29.6	21.4	27.9	20.2	26.6	19.2	22.8	27.8	21.6	26.5	20.4	24.8	4.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	16.5	26.0	19.9	15.2	24.6	18.8	14.1	23.3	69.1	27.9	64.5	26.6	60.4	24.8	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
9.7	8.5	7.7	DB	-35.1	32.2	4.1	1.2	-38.0	33.1	-40.4	33.8	-42.7	34.5	-45.6	35.4	
			WB	-33.7	24.7	2.7	1.1	-35.6	25.5	-37.2	26.2	-38.7	26.8	-40.6	27.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.8	-14.5	-12.0	-4.6	4.1	11.1	16.8	19.6	18.3	13.5	6.1	-2.8	-11.1
	DBStd	13.26	8.24	7.74	7.65	5.66	4.65	3.69	3.00	3.19	4.49	4.80	6.39	7.51
	HDD10.0	3257	758	616	454	188	43	2	0	0	16	142	384	654
	HDD18.3	5447	1016	849	711	428	229	70	22	41	154	381	634	912
	CDD10.0	991	0	0	2	11	76	207	298	257	122	19	1	0
	CDD18.3	138	0	0	0	0	3	24	61	39	10	1	0	0
	CDH23.3	1370	0	0	1	5	60	260	561	371	105	8	0	0
	CDH26.7	302	0	0	0	1	11	57	132	80	21	1	0	0
Wind	WSAvg	3.4	3.4	3.4	3.5	3.9	3.8	3.3	3.0	3.0	3.3	3.7	3.6	3.4
Precipitation	PrecAvg	572	14	9	18	34	68	101	87	85	67	51	23	15
	PrecMax	762	67	32	90	99	154	368	164	199	230	193	95	44
	PrecMin	369	1	0	0	0	12	22	28	18	9	5	1	0
	PrecStd	102	13	7	16	24	39	58	31	46	44	39	19	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.6	6.1	17.7	23.4	28.3	31.2	31.9	31.1	29.1	23.9	15.2	4.6
		MCWB	2.0	3.0	11.9	13.0	17.5	22.5	22.9	22.7	20.6	15.6	10.2	1.9
	2%	DB	1.4	3.6	11.5	19.6	25.0	28.0	29.7	28.8	26.4	19.9	11.0	2.2
		MCWB	0.0	1.4	6.9	10.2	15.2	19.8	21.7	20.9	18.8	13.4	7.1	0.6
	5%	DB	-0.5	1.7	7.7	16.6	22.5	26.3	28.0	27.2	23.8	16.7	7.9	0.8
		MCWB	-1.7	0.3	4.1	8.8	14.0	18.4	20.7	19.9	17.5	11.6	5.1	-0.3
	10%	DB	-2.8	-0.4	5.1	13.8	20.1	24.4	26.6	25.5	21.7	13.6	5.4	-1.0
		MCWB	-3.7	-1.6	2.4	7.3	12.9	17.6	19.8	18.7	16.5	9.6	3.4	-2.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.2	3.6	12.2	13.5	19.7	23.6	24.4	24.0	22.2	16.9	10.8	2.4
		MCDB	4.2	5.4	14.6	20.7	24.9	28.5	29.6	29.2	27.7	20.7	13.7	3.7
	2%	WB	0.3	1.7	7.2	11.1	17.2	21.5	23.0	22.3	20.2	14.5	7.6	0.9
		MCDB	1.4	3.3	11.1	17.5	21.7	27.0	27.7	27.2	24.1	19.6	10.2	1.8
	5%	WB	-1.6	0.3	4.4	9.6	15.4	19.9	21.9	20.9	18.6	11.9	5.4	-0.4
		MCDB	-0.5	1.7	7.3	15.5	20.1	24.2	26.7	25.3	22.6	15.5	7.7	0.7
	10%	WB	-3.7	-1.6	2.5	7.9	13.8	18.6	20.7	19.8	17.0	10.0	3.3	-2.0
		MCDB	-2.8	-0.4	4.9	13.6	19.0	22.7	25.0	24.1	20.7	13.2	5.3	-1.1
Mean Daily Temperature Range	5% DB	MDBR	10.1	11.1	11.2	11.4	11.7	11.1	11.6	11.8	11.1	9.1	7.8	8.4
		MCDBR	9.7	10.7	12.6	16.6	15.9	14.2	13.4	13.8	14.5	14.5	11.6	8.8
		MCWBR	8.5	8.6	8.4	9.5	8.1	7.2	6.9	7.1	7.9	8.7	7.9	7.5
	5% WB	MCDBR	9.4	9.4	12.0	14.8	12.9	11.9	12.0	12.0	13.3	10.6	8.4	
		MCWBR	8.4	7.7	8.3	9.1	7.7	7.0	7.1	6.8	7.3	8.4	7.6	7.4
Clear-Sky Solar Irradiance	taub	0.250	0.268	0.305	0.349	0.358	0.375	0.380	0.375	0.351	0.305	0.278	0.253	
	taud	2.352	2.309	2.372	2.370	2.409	2.386	2.394	2.427	2.475	2.575	2.499	2.386	
	Ebn at Noon	833	894	913	900	903	885	875	864	854	850	798	781	
	Edh at Noon	66	88	100	111	112	115	113	105	90	69	59	57	
All-Sky Solar Radiation	RadAvg	1.40	2.61	3.80	4.79	5.38	5.66	6.07	5.06	3.75	2.21	1.25	1.01	
	RadStd	0.15	0.21	0.38	0.42	0.33	0.44	0.28	0.37	0.30	0.32	0.14	0.11	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	-1.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page